

GeoSnap : Land-Use Classification and Explainability from Space

Technical Report

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Abstract

Satellite imagery is widely used for land-use monitoring, environmental assessment, and geospatial planning. This work investigates land-use classification on the EuroSAT dataset using two modalities: RGB imagery and 13-band Sentinel-2 multispectral imagery. ResNet18-based convolutional neural networks were trained for both approaches to compare the effectiveness of visual and spectral information.

Results show that the multispectral model outperforms the RGB baseline, particularly for spectrally similar vegetation classes. Model behavior was analyzed using confusion matrices, t-SNE feature visualization, Grad-CAM, spectral signature analysis, and band importance evaluation. The findings highlight the importance of Near-Infrared (NIR), Red Edge, and Shortwave Infrared (SWIR) bands for accurate land-cover discrimination.

Environmental insights were further derived using NDVI, NDMI, NDWI, NDBI, and NDRE indices, revealing clear separation between vegetation, water, and built-up classes. Overall, the study demonstrates that multispectral imagery improves both classification performance and interpretability, highlighting the value of spectral information for Earth observation applications.

1 Introduction

Satellite-based land-use classification is a foundational task in geospatial intelligence, enabling applications ranging from agricultural monitoring and urban planning to environmental protection and disaster response. The EuroSAT dataset, derived from the European Space Agency’s Sentinel-2 satellite, provides a standardized benchmark of 64×64 pixel image patches across ten land-use classes, available in both three-band RGB and full 13-band multispectral modalities. This report investigates the extent to which spectral information beyond the visible range improves classification accuracy and class separability, with particular attention to classes that appear visually similar in RGB but are spectrally distinguishable using near-infrared and shortwave infrared bands. Beyond accuracy, the report emphasizes model interpretability: understanding which spatial regions and spectral bands drive predictions is essential for trust and deployment in real-world geospatial systems.

2 System Architecture

The proposed system consists of four main stages: data preprocessing, feature extraction, classification, and model interpretation. Sentinel-2 multispectral image patches are first loaded and normalized before being provided as input to a modified ResNet-18 network. The network extracts hierarchical spectral-spatial features through a series of convolutional and residual layers, producing a 512-dimensional feature representation for each image. These features are used by a fully connected classification layer to predict one of the ten EuroSAT land-cover classes. Finally, interpretability and analysis modules, including confusion matrix evaluation, t-SNE visualization, band ablation studies, spectral signature analysis, and Grad-CAM, are employed to understand the model’s decision-making process and feature utilization.

3 Methodology

This study investigates land-use and land-cover classification using RGB and multispectral Sentinel-2 imagery from the EuroSAT benchmark dataset. Two separate deep learning pipelines were developed and evaluated: (i) an RGB-based classification model utilizing three visible spectral bands and (ii) a multispectral classification model utilizing all thirteen Sentinel-2 bands. Both approaches were implemented using the ResNet18 convolutional neural network architecture and trained using the official EuroSAT train-validation split provided. Both models were trained using the Cross Entropy Loss function, which is suitable for multi-class image classification problems. The Adam optimizer was used with a learning rate of 1×10^{-3} for the RGB model and 1×10^{-4} for the multispectral model. For the RGB model, a weight decay of 1×10^{-4} was additionally applied to improve generalization.

The models were initialized using ImageNet-pretrained ResNet18 weights. The RGB model retained the original 3-channel input layer, whereas the multispectral model modified the first convolutional layer to accept 13 Sentinel-2 bands. The pretrained RGB filters were averaged and replicated across all spectral channels, allowing the network to benefit from transfer learning while adapting to multispectral inputs.

Training was carried out for 10 epochs for the RGB model and 20 for the multispectral model, both with a batch size of 32. During each epoch, the model processed batches of training images, computed the classification loss, performed backpropagation, and updated the network weights. After every epoch, the model was evaluated on the validation dataset, and the model achieving the highest validation accuracy was saved as the final trained network. To evaluate the effectiveness of the proposed approach, training accuracy, validation accuracy, training loss, and validation loss were monitored throughout the learning process.

3.1 RGB Classification Model

The RGB classification model utilizes only the three visible Sentinel-2 bands (Red, Green, and Blue) as input to the ResNet18 architecture. Since the input consists of three channels, the original ImageNet-pretrained ResNet18 model was used without modifying the first convolutional layer. Transfer learning enabled the model to leverage pretrained visual features, resulting in faster convergence and improved classification performance.

Prior to training, several preprocessing and data augmentation techniques were applied using the torchvision.transforms library. Each training image was resized to 224×224 pixels to match the input size required by the ResNet18 architecture, randomly flipped horizontally, randomly rotated by up to 10° , converted into a tensor, and normalized using the standard ImageNet mean and standard deviation values. These augmentation techniques improve the model’s ability to generalize by introducing variations in image orientation while reducing the risk of overfitting. For validation and testing, only resizing and normalization were performed to ensure consistent evaluation. Instead of training the network from scratch, the final fully connected classification layer of the pre-trained model was replaced with a new linear layer consisting of ten output neurons corresponding to the ten EuroSAT classes. The remaining layers retained their learned feature representations and were fine-tuned during training, enabling the network to adapt efficiently to satellite imagery.

3.2 Multispectral Classification Model

To accommodate multispectral imagery, the first convolutional layer was modified from 3 input channels to 13 channels, with the pretrained RGB weights averaged and replicated across channels to provide a stable initialization rather than random weights. Before training, the Sentinel-2 band ordering was independently verified rather than assumed. This was done through NDVI sanity checks, resolution-based sharpness analysis showing the expected lower texture variance in the 60 m bands (B1 and B9), and cross-referencing the official Sentinel-2 bandId convention. This verification step ensured that spectral information was mapped correctly to the network input channels. Although ResNet-18 was originally designed for 224×224 inputs, the 64×64 native resolution of EuroSAT patches was used directly for multispectral training. The global average pooling layer accommodates this reduced spatial resolution without architectural changes, though the final feature maps are compressed to 2×2 spatial resolution before pooling. The network accepted $13 \times 64 \times 64$ input and began with a 7×7 convolution that produced 64 feature maps, followed by batch normalization, ReLU activation, and maximum pooling. Four residual stages progressively learned increasingly complex spectral-spatial representations: early layers captured local textures and spectral gradients, intermediate layers

combined these into larger structures such as roads, fields, and vegetation patterns, while deeper layers formed class-level representations. The final layer4 featuremaps were globally averaged and pooled into a 512-dimensional feature vector, which was subsequently used both for classification through the final fully connected layer and for feature-space visualization using t-SNE.

4 Results

The RGB ResNet18 model converged successfully, achieving a final training accuracy of 93.02% and a best validation accuracy of 93.43%. On the test dataset, the model achieved an overall accuracy of 93.00%. The classification report showed balanced performance with macro precision, recall, and F1-score of approximately 0.93.

The multispectral ResNet-18 model achieved a training accuracy of 99.6% and a validation accuracy of 96.6% with macro precision, recall, and weighted F1-score of 0.97.

5 Explainability Analysis for RGB Classification

5.1 Confusion Matrix Analysis

The confusion matrix provides a detailed evaluation of the RGB ResNet18 model by illustrating the relationship between the true class labels and the corresponding predicted labels. Unlike the overall accuracy metric, the confusion matrix enables a class-wise assessment of the model’s performance and highlights the specific categories where misclassifications occur.

The generated confusion matrix shows a strong concentration of values along the main diagonal, indicating that the majority of satellite images are correctly classified into their respective land-cover categories. This demonstrates that the model has successfully learned highly discriminative visual representations for most classes in the EuroSAT dataset. The consistently high number of true positive predictions across multiple classes reflects the effectiveness of transfer learning with ResNet18 in extracting meaningful spatial and spectral features from RGB satellite imagery.

The off-diagonal elements of the confusion matrix correspond to incorrect predictions. These misclassifications are relatively infrequent and mainly occur between land-cover classes that exhibit similar visual characteristics. For instance, categories containing comparable vegetation density, built-up structures, or agricultural patterns may occasionally be confused because of overlapping textures, colors, and spatial arrangements in aerial imagery. Such errors are expected in remote sensing applications, where different land-cover types can share similar appearances when viewed from above.

Despite these occasional confusions, the overall distribution of predictions indicates that classification errors are minimal and do not significantly affect the model’s overall performance. The confusion matrix therefore confirms that the RGB ResNet18 model achieves balanced performance across different classes without exhibiting substantial bias toward any particular category. This analysis supports the quantitative evaluation metrics and demonstrates the robustness and reliability of the proposed model for satellite image scene classification.

5.2 Grad-CAM Visualization

To understand the decision-making process of the RGB ResNet18 model, Gradient-weighted Class Activation Mapping (Grad-CAM) was employed. Grad-CAM generates heatmaps that highlight the image regions contributing most to the model’s predictions, thereby improving the interpretability of the model.

The generated visualizations indicate that the model focuses primarily on the most discriminative regions of the satellite images. For example, in residential scenes, attention is concentrated on building clusters and road networks, while in forest and river scenes, the model emphasizes dense vegetation and water bodies. These highlighted regions align well with the distinguishing characteristics of their respective classes.

Overall, the Grad-CAM analysis suggests that the RGB ResNet18 model relies on meaningful spatial features rather than irrelevant background information, increasing confidence in the model’s classification decisions.

5.3 Error Analysis

To further evaluate the model, a class-wise error analysis was performed using the confusion matrix. For each land-cover category, the numbers of correct and incorrect predictions were calculated along with the corresponding classification accuracy.

The analysis shows that most classes achieve high classification accuracy, indicating that the model effectively learns the distinguishing visual characteristics of different land-cover types. The few errors observed are primarily associated with classes that exhibit similar textures or spatial patterns in satellite imagery, leading to occasional confusion.

Overall, the low number of misclassifications demonstrates that the RGB ResNet18 model generalizes well on the EuroSAT dataset and provides reliable predictions across different scene categories.

6 Explainability Analysis of Multispectral Classification Model

6.1 Spectral Signature Analysis

To understand class separability, average spectral signatures were computed for all ten land-cover classes. The spectral profiles reveal clear differences in reflectance behavior across vegetation, water, and built-up categories.

Vegetation classes such as Forest, Pasture, AnnualCrop, PermanentCrop, and HerbaceousVegetation exhibit a pronounced increase in reflectance in the Red Edge and Near-Infrared bands (B5–B8), which is characteristic of healthy vegetation due to strong NIR reflection.

In contrast, water classes, especially SeaLake, maintain very low reflectance across both NIR and SWIR bands, producing a highly distinctive spectral profile that is fundamentally different from vegetation and urban surfaces. This strong spectral separability explains the near-perfect classification accuracy achieved for SeaLake.

Built-up classes including Residential, Industrial, and Highway exhibit flatter spectral responses that generally lie between vegetation and water signatures.

The greatest inter-class separability occurs in the NIR (B8) and SWIR (B11–B12) bands, indicating that these wavelengths provide the most discriminative information for land-cover classification.

6.2 Feature Space Visualization

The 512-dimensional features learned by the network were projected into two dimensions using t-SNE. The axes themselves do not have physical meaning; instead, the relative distances between points indicate similarity in the learned feature space. The resulting visualization revealed distinct clusters for most classes, indicating that the network learned meaningful and discriminative feature representations. Classes such as SeaLake, River, and Forest formed well-separated clusters, while vegetation-related categories occupied nearby regions, reflecting their similar spectral characteristics and explaining the confusion observed among these classes.

6.3 Confusion Matrix Analysis

The confusion matrix reveals that nearly all classification errors occur within a single cluster of vegetation-dominated land-cover types, while built-up and water classes remain almost entirely free of confusion. The most prominent error involves PermanentCrop, which was misclassified as HerbaceousVegetation in 15 cases and as AnnualCrop in 11 cases. A secondary pattern appears with Pasture, which was confused with HerbaceousVegetation in 9 cases. By contrast, classes such as SeaLake, Forest, and Residential show negligible cross-class error, consistent with their highly distinctive spectral signatures. This pattern is well explained by the underlying spectral behavior of these land-cover types. AnnualCrop, PermanentCrop, Pasture, and HerbaceousVegetation all share the defining characteristic of healthy green vegetation: strong chlorophyll absorption in the red band and strong reflectance in the near-infrared, producing comparably high NDVI values across all four classes. The features that genuinely separate them are subtler and tied to canopy structure and moisture rather than simple greenness. PermanentCrop, typically orchard or vineyard cover, exhibits gaps of exposed soil between plants, which raises red and SWIR reflectance relative to a continuous canopy and slightly lowers NDVI compared to forest-like vegetation; this soil-gap signal is the primary cue that should distinguish it from denser annual cropland or natural grassland, but it is a second-order effect that a global average spectral signature per class does not fully capture. Pasture, meanwhile, tends to have lower biomass and drier ground cover than irrigated cropland, producing elevated SWIR reflectance relative to AnnualCrop even at similar NIR levels. HerbaceousVegetation occupies a similar spectral space to Pasture, differing mainly in moisture content and seasonal state rather than any single dominant band. Because the model receives only a single-timestamp image, it cannot observe the temporal patterns, such as crop rotation cycles, irrigation timing, or mowing frequency, that would more reliably separate these classes in practice. The confusion matrix therefore reflects a genuine spectral ambiguity inherent to single-date multispectral imagery rather than a deficiency in the model architecture itself. This finding is independently corroborated by the t-SNE feature visualization, where PermanentCrop and HerbaceousVegetation show partial overlap in the learned 512-dimensional feature space, while spectrally unambiguous classes such as SeaLake and Industrial form tight, well-isolated clusters.

6.4 Band Ablation Analysis

To quantify the contribution of each spectral band to model performance, a permutation-based ablation analysis was performed. Each of the 13 Sentinel-2 bands was individually zeroed out in the validation set, and the resulting decrease in classification accuracy was measured relative to the unablated baseline accuracy of 96.6%.

Among the visible and near-infrared bands, B4 (Red) produced the largest accuracy drop, consistent with its central role in chlorophyll absorption and its importance in vegetation-sensitive indices. B2 (Blue) and B3 (Green) also showed substantial contributions, reflecting their importance for general land-cover discrimination across the visible spectrum.

The atmospheric band B9 (Water Vapor) produced an accuracy drop that was effectively zero, indicating that it contributes little to land-cover classification. This result is consistent with remote-sensing theory, as B9 is designed to measure atmospheric water vapor rather than surface reflectance.

An unexpected result was observed for B10 (Cirrus), which produced the largest accuracy drop of any individual band. Since B10 is intended primarily for cloud and haze detection rather than surface characterization, this finding should be interpreted cautiously. One possible explanation is that the band captures acquisition-specific atmospheric conditions that are indirectly correlated with certain classes or regions in the dataset. Another possibility is that the zero-ablation procedure introduces an out-of-distribution input that the network never encountered during training, causing the measured importance to be artificially inflated. Consequently, the result should be viewed as an interesting empirical observation rather than evidence that cirrus information directly discriminates land-cover classes.

The SWIR bands, B11 and B12, exhibited moderate importance, consistent with their established role in moisture-sensitive and built-up area indices such as NDMI and NDBI. Vegetated surfaces generally display SWIR responses influenced by canopy moisture content, whereas built-up and bare-soil surfaces often exhibit stronger reflectance in this region of the spectrum. Overall, the ablation results indicate that the model relies most heavily on a combination of visible, NIR, and SWIR information, which aligns closely with the spectral-signature analysis and the known spectral characteristics of vegetation, water, and urban land-cover types.

6.5 Grad-CAM Visualization

To assess which spatial regions of each image most influenced the model’s predictions, Grad-CAM was applied to the final convolutional block (layer4) of the trained ResNet-18, producing class-discriminative heatmaps overlaid on the input imagery. For correctly classified examples, such as an AnnualCrop patch predicted with high confidence, the model’s attention was distributed broadly and relatively uniformly across the image rather than concentrated on a single localized region. This behavior is consistent with the spatial homogeneity of cropland, where the discriminative signal is the overall spectral-textural character of the patch rather than any single distinguishing feature.

This pattern contrasts sharply with classes possessing a clear geometric structure. Grad-CAM was applied to a correctly classified Highway patch, where the road crossing the image forms a visible diagonal structure. Here, attention concentrates specifically along this linear pavement feature rather than spreading across the patch, with peak activation aligned to the road itself. This demonstrates that the model adapts its spatial

attention strategy to whichever cue is most discriminative for a given class: diffuse textural signals for vegetation-dominated classes, and localized structural features for classes like Highway where a single, well-defined geometric pattern is the primary distinguishing characteristic.

To directly probe a known failure mode, Grad-CAM was also applied to a correct and an incorrect prediction within the PermanentCrop class, the category most frequently confused with HerbaceousVegetation in the confusion matrix. In the correctly classified example, the image shows a clear row or plot structure characteristic of orchard or vineyard cultivation, and the model’s attention concentrates over the region where this structure is most visually distinct. In the misclassified example, predicted as HerbaceousVegetation, this structural cue is largely absent; the patch instead shows a patchy, mottled texture without clear geometric regularity, and the model’s attention settles on a similarly irregular region. This supports the explanation proposed that PermanentCrop’s spectral and structural distinctiveness depends substantially on visible row or soil-gap patterns between plants, and when this structural signal is faint or absent in a given sample, the model defaults toward the visually and spectrally closer HerbaceousVegetation class.

Similar attention patterns were observed for Pasture and Residential samples, where the model focused on vegetation-dominated regions and urban structural features respectively, further confirming that predictions were driven by meaningful land-cover characteristics.

7 Environmental Insights

To extract insights into ecological and urban-planning beyond the classification accuracy, five spectral indices were calculated for each validation image: NDVI, NDWI, NDMI, NDBI, and NDRE. These five spectral indices were selected because they capture the primary physical properties that differentiate the EuroSAT land-cover classes. NDVI measures vegetation greenness and biomass, NDMI captures vegetation moisture content, NDWI identifies water bodies, NDBI highlights built-up and impervious surfaces, and NDRE exploits Sentinel-2’s red-edge bands to quantify vegetation vigor and chlorophyll content.

The mean index profiles across all ten classes, shown in the heatmap, reveal three clearly distinct spectral regimes. The natural vegetation group (Forest, Pasture, HerbaceousVegetation, AnnualCrop, PermanentCrop) shows strongly positive NDVI and NDMI, with Forest registering the highest values across all vegetation indices, consistent with a dense, moisture-rich closed canopy with high chlorophyll content. Pasture shows the second highest NDMI, but lower than Forest, reflecting the faster moisture loss of grass cover relative to tree canopy. AnnualCrop and PermanentCrop occupy a similar spectral range despite being distinct land-use classes, which directly explains the confusion matrix errors and their index profiles are nearly indistinguishable from mean reflectance alone.

The water group (SeaLake, River) is uniquely identifiable through its strongly negative NDVI and high NDMI, the latter reflecting the fact that water absorbs SWIR far more strongly than NIR, driving the NDMI numerator positive. The NDWI index correctly flags SeaLake as the only class with a positive mean value, confirming water presence. The built-up group (Residential, Industrial, Highway) shows uniformly low NDVI and weakly negative NDBI, with Industrial recording the lowest vegetation signal of any land class. Highway shows a somewhat higher NDVI than expected for asphalt, likely because

roadside vegetation and surrounding fields contribute to the 64×64 patch mean. This highlights a known limitation of patch-level analysis: spectrally mixed patches reduce index discriminability for linear features like roads.

The NDVI–NDBI scatter plot provides the clearest link between the multispectral analysis and potential land-use planning applications. A pronounced negative relationship is observed across the 4,050 validation samples, with vegetation-dominated classes occupying the high-NDVI, low-NDBI region and built-up classes clustering toward lower NDVI and higher NDBI values. Water bodies form a distinct and well-separated group characterized by negative NDVI and strongly negative NDBI values. This clear separation demonstrates that vegetation abundance and built-up intensity represent two complementary dimensions for describing land-cover composition.

Beyond classification, the NDVI–NDBI feature space offers a useful framework for environmental monitoring and land suitability assessment. In a georeferenced setting, vegetation-rich areas located near regions exhibiting elevated built-up intensity could be identified as zones experiencing urban expansion pressure. Such spectral relationships are widely used in green-space monitoring, urban-growth assessment, and sustainable land-use planning, highlighting the practical value of multispectral remote sensing beyond scene classification alone.

8 Conclusion

This project compared RGB and multispectral ResNet18 models for EuroSAT land-use classification and showed that incorporating the full Sentinel-2 spectrum improves performance and interpretability. The RGB model provided a strong baseline using only visible information, while the multispectral model achieved better overall results by exploiting additional Red Edge, NIR, and SWIR bands. The improvement was most noticeable for vegetation-related classes, where spectral similarity in RGB led to confusion, but multispectral data provided clearer separation through moisture, chlorophyll, and canopy structure information.

The analysis also confirmed that the model learned physically meaningful representations rather than relying on irrelevant image regions. Spectral signatures, band ablation, t-SNE, and Grad-CAM all showed that the network focused on class-relevant spectral and spatial cues. At the same time, some classes such as AnnualCrop, PermanentCrop, Pasture, and HerbaceousVegetation remained difficult to separate because they share similar single-date spectral characteristics. This shows that multispectral imagery improves class discrimination, but temporal variation is still important for fully resolving spectrally similar vegetation classes.

Future work could extend this study in several directions. First, using multi-date or time-series imagery would help capture seasonal changes, crop rotation, mowing cycles, and harvest patterns that cannot be observed in a single snapshot. Second, additional spectral indices and geospatial context could be incorporated for tasks such as land suitability assessment, urban expansion monitoring, and vegetation change detection. Third, experimenting with more advanced architectures, stronger augmentation, or class-balanced training may further improve performance on the most confused vegetation classes. Finally, combining multispectral classification with spatially contiguous geographic data would make the system more useful for real-world environmental monitoring and planning applications.

A Dataset Details

- Dataset : EuroSAT Dataset
- Dataset Modalities : RGB (3-band) and Multispectral (13-band)
- Number of classes : 10
- Classes : AnnualCrop, Forest, HerbaceousVegetation, Highway, Industrial, Pasture, PermanentCrop, Residential, River, and SeaLake.
- Band Order : B1, B2, B3, B4, B5, B6, B7, B8, B8A, B9, B10, B11, B12

B System Parameters

Table 1: Model Training Parameters

Parameter	RGB Model	Multispectral Model
Backbone	ResNet18	ResNet18
Pretraining	ImageNet	ImageNet
Input Size	224×224	64×64
Input Channels	3	13
Epochs	10	20
Batch Size	32	32
Optimizer	Adam	Adam
Learning Rate	0.001	0.0001
Weight Decay	1×10^{-4}	0
Loss Function	CrossEntropyLoss	CrossEntropyLoss
Preprocessing	Augmentation: Random horizontal flip, random rotation ($\pm 10^\circ$)	Normalization: Reflectance values divided by 5000

C Spectral Indices

Index	Full Name	Formula
NDVI	Normalized Difference Vegetation Index	$\frac{B8-B4}{B8+B4}$
NDWI	Normalized Difference Water Index	$\frac{B3-B8}{B3+B8}$
NDMI	Normalized Difference Moisture Index	$\frac{B8-B11}{B8+B11}$
NDBI	Normalized Difference Built-up Index	$\frac{B11-B8}{B11+B8}$
NDRE	Normalized Difference Red Edge Index	$\frac{B8-B5}{B8+B5}$